

Total No. of Questions : 4]

SEAT No. :

**PF282**

[Total No. of Pages : 2

**Apr-26/TE/Insem-361**

**T.E. (Civil Engineering) (Insem)**

**WASTE WATER ENGINEERING**

**(2019 Pattern) (Semester - VI) (301012)**

**Time : 1 Hour]**

**[Max. Marks : 30**

**Instructions to the candidates:**

- 1) Attempt Q1 or Q2 and Q3 or Q4.
- 2) Neat diagrams must be drawn whenever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary and clearly state.
- 5) Use of electronic pocket calculator is allowed.

**Q1) a)** Write factors considered while estimating sewage flow. Determine the domestic sewage flow generated from a town using following data, Population: 80000 and rate of water supply: 150 lpcd of which 80% contributes to sewage generation. Consider Peak factor 2. **[2+3]**

b) Differentiate between separate sewerage system and combined sewerage system? **[5]**

c) Design a circular sewer for the conveyance 5 MLD of sewage flow. The sewer should be designed to carry maximum discharge while running half full. A velocity of 1.0-1.5 m/s should be generated at maximum flow. Use the following data. **[5]**

i) Max. flow/Ave. flow = 3

ii) Manning's constant = 0.013

OR

**Q2) a)** Enumerate the causes to consider the emerging contaminants of waste water while designing a sewage treatment plant. Write any 3 emerging contaminants in waste water. **[5]**

b) Explain the various zones of pollution for a polluted river undergoing self-purification. **[4]**

c) The 5 day BOD of a wastewater sample at 20°C was found to be 300 mg/l. Determine the ultimate BOD and 3day BOD at 30°C. Take  $K_D = 0.1/d$  at 20°C. **[6]**

**P.T.O.**

- Q3)** a) Draw a flowchart for sewage treatment plant consisting primary and secondary treatment. Enlist the pollutants removed from each unit operation and process. [5]
- b) Design a grit chamber for the following data : [5]
- Flow =  $1500 \text{ m}^3/\text{hour}$
  - Setting velocity of particle  $0.01 \text{ m/s}$
  - Flow through velocity  $0.3 \text{ m/s}$
  - Depth of grit chamber  $0.5 \text{ m}$
- c) Design primary settling tank of rectangular shape for a town having a population of 130000 with sewage generation rate of 150 lpcd. Surface overflow rate  $50 \text{ m}^3/\text{d}/\text{m}^2$ . Consider detention period = 2hrs. Assume length to width ratio as 2 : 1. [5]

OR

- Q4)** a) Write the significance of provision of equalization, neutralization and skimming tanks in effluent treatment plant. [3]
- b) Find number of openings and number of bars of a screen to handle 25 MLD of waste with approach velocity  $1 \text{ m/s}$ . Assume depth of flow =  $0.9 \text{ m}$ , size of each opening =  $25 \text{ mm}$ , diameter of bar =  $10 \text{ mm}$ . Also find head loss to screen if screen is kept at an angle of  $60^\circ$  with horizontal. [6]
- c) Determine diameter and depth of primary sedimentation tank for sewage flow 30 million litres per day. [6]

Given data :

- Detention time = 2.5 hours
- Surface loading rate =  $40000 \text{ l}/\text{m}^2/\text{d}$
- Freeboard  $0.5 \text{ m}$  and sludge accumulation space  $0.5 \text{ m}$ .

x

x

x